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For example

```
int main()
{


    int i=0;
    while(++i < 10000 )
                                  This is an empty loop because the while
                                  loop does not contain any statement
    return 0;

}
                    }
```

In the above code, the loop is a time delay loop. A time delay loop is useful for pausing the program for some time.

A while loop may be infinite loop when no update statement inside the body of the loop. For example,

```
int main()
{

    int i = 0;
    while(i < =10)
      cout <<"The value of i is "<<i;   -->  This statement is infinitely displaying because no update statement inside body of the loop
      i++;
    return 0;                            -->  This is not part of the while loop statementt because of missing curly braces

}
```

Similarly, there is another variation of while is also shown below:

```
int main()
{
    int i=1;
    while( ++i  < 10)
    cout<< "The value of i is "<<i;
    return 0;
}
```